

Supplementary material for Nonlinear Shrinkage Estimation of Large Integrated Covariance Matrices

1. MARKET TRADE SIMULATIONS

We simulate intra-day price data and construct minimum variance portfolio for each of the seven methods compared in Section 4.2. 5

Following Barndorff-Nielsen et al. (2011) and Fan et al. (2012), we simulate $p = 100$ and $p = 200$ stock prices for 210 days, which is 42 weeks of trading days. We use $X_t^{o(i)} = X_t^{(i)} + \epsilon_t^{(i)}$, where $X_t^{(i)}$ is the underlying log-price, and $\epsilon_t^{(i)}$ models the market microstructure noise, with $\epsilon_t^{(i)} \sim N(0, 0.0005^2)$ and are assumed to be independent of each other. The log-price $X_t^{(i)}$ is generated by 10

$$dX_t^{(i)} = \mu^{(i)} dt + \rho^{(i)} \sigma_t^{(i)} dB_t^{(i)} + \{1 - (\rho^{(i)})^2\}^{1/2} \sigma_t^{(i)} dW_t + \nu^{(i)} dZ_t \quad (i = 1, \dots, 100),$$

where $\{W_t\}$, $\{Z_t\}$ and the $\{B_t^{(i)}\}$'s are all independent standard Brownian motions. The processes $\{Z_t\}$ and $\{W_t\}$ play the role of factors, when a real market usually has a market factor in the asset returns. The spot volatility $\sigma_t^{(i)} = \exp(\varrho_t^{(i)})$ follows the independent Ornstein-Uhlenbeck process

$$d\varrho_t^{(i)} = \alpha^{(i)}(\beta_0^{(i)} - \varrho_t^{(i)})dt + \beta_1^{(i)} dU_t^{(i)},$$

where the $\{U_t^{(i)}\}$'s are independent standard Brownian motions. We use $(\mu^{(i)}, \beta_0^{(i)}, \beta_1^{(i)}, \alpha^{(i)}, \rho^{(i)}) = (0.03x_1^{(i)}, -x_2^{(i)}, 0.75x_3^{(i)}, -1/40x_4^{(i)}, -0.7)$ and $\nu^{(i)} = \exp(\beta_0^{(i)})$, where the $x_j^{(i)}$'s are independent and uniformly distributed on the interval $[0.7, 1.3]$. The initial value of each log-price is set at $X_0^{(i)} = 1$ and the starting spot volatility $\varrho_0^{(i)} = 0$. 15

We simulate the trading times independently from the price data assuming the transaction times for each stock follow independent Poisson processes with rates $\lambda_1, \dots, \lambda_{100}$ respectively, where $\lambda_i = 0.01i(23400)$ ($i = 1, \dots, 100$), because normal trading time for one day has 23400 seconds. 20

After simulating the data, we split a trading day into 15-minute intervals, and set the price data for each stock at the end of each interval as the price observed at the trade right before the end of the interval. The data is used to calculate various integrated covariance matrix estimators. At the start, we invest 1 unit of capital using (6) constructed from different estimators of Σ_p . We re-evaluate the portfolio weights every week, using either the past 8 weeks or 4 weeks of data as the training set. To gauge performances, we compute the annualized return $\hat{\mu}$ and the annualized out-of-sample standard deviation $\hat{\sigma}$, defined similarly as those in Section 4.2, together with the Sharpe ratio $\hat{\mu}/\hat{\sigma}$ and the actual risk of portfolios. 25

From Table 1, all methods, except for the realized covariance, are similar in terms of the actual risk. The annualized returns are higher for all methods when using 8-week training windows instead of 4, while the annualized out-of-sample standard deviation or the actual risk do not change much. The maximum exposure level also decreased in general as we change from 4-week training windows to 8-week. Overall, all measures from all methods are not too different from one another. The results in Section 4.2 in the paper are quite different. 30

Table 1. *Market trades simulation results. Standard errors for weekly maximum exposures are subscripted. Graphical lasso is omitted because of non-convergence issues.*

	$p = 100$					$p = 200$		
	Annualized Return	Annualized Std. Dev.	Sharpe Ratio	Maximum Exposure (%)	Actual Risk	Sharpe Ratio	Maximum Exposure (%)	Actual Risk
Weekly rebalancing with 4-week training windows								
Theoretical	14.6	15.5	0.94	3.3 _{1.2}	13.1	0.91	2.9 _{1.0}	13.1
RCV	23.1	15.4	1.50	14.4 _{5.3}	14.5	0.92	18.2 _{6.8}	16.6
Proposed	21.0	15.3	1.37	7.5 _{2.6}	13.4	0.98	9.1 _{3.8}	13.6
Banding	23.2	15.2	1.53	8.3 _{3.0}	13.6	1.12	9.8 _{3.8}	14.2
Grand Avg	21.1	15.2	1.38	7.5 _{2.6}	13.4	0.98	9.2 _{3.8}	13.6
NONLIN	21.1	15.2	1.39	7.4 _{2.6}	13.4	1.03	8.7 _{3.6}	13.6
POET	21.2	15.3	1.38	7.6 _{2.7}	13.4	1.03	7.9 _{3.2}	13.6
SCAD	21.1	15.0	1.41	8.1 _{3.0}	13.5	1.20	10.8 _{4.1}	14.3
Weekly rebalancing with 8-week training windows								
Theoretical	21.2	15.8	1.34	2.4 _{1.2}	13.1	1.56	2.2 _{1.1}	13.1
RCV	25.9	15.6	1.66	10.0 _{5.1}	13.8	1.14	12.7 _{6.9}	14.6
Proposed	23.1	15.6	1.48	6.1 _{3.1}	13.2	1.66	8.6 _{4.7}	13.4
Banding	24.5	15.8	1.55	5.8 _{2.9}	13.3	1.65	7.2 _{3.9}	13.5
Grand Avg	22.5	15.6	1.44	6.2 _{3.1}	13.2	1.65	8.6 _{4.7}	13.4
NONLIN	22.8	15.7	1.46	6.0 _{3.0}	13.2	1.65	8.0 _{4.5}	13.4
POET	22.7	15.6	1.45	5.6 _{2.8}	13.3	1.61	6.8 _{3.7}	13.4
SCAD	24.2	15.8	1.53	5.6 _{2.8}	13.3	1.65	7.8 _{4.3}	13.5

RCV, realized covariance; Grand Avg, grand average; NONLIN, nonlinear shrinkage; POET, principal orthogonal complement thresholding; SCAD, adaptive thresholding with the smoothly clipped absolute deviation penalty.

2. PROOF OF RESULTS

Proof of Lemma 1. Since M is finite, if we can show

$$\|\text{diag}(P_1^{(j)\top} \tilde{\Phi}_2^{(j)} P_1^{(j)}) \text{diag}^{-1}(P_1^{(j)\top} \Phi P_1^{(j)}) - 1\| \rightarrow 0$$

almost surely for $j = 1$, then the result of the lemma is established. Dropping the superscript (j) in all notations, we are proving

$$\|\text{diag}(P_1^\top \tilde{\Phi}_2 P_1) \text{diag}^{-1}(P_1^\top \Phi P_1) - 1\| \rightarrow 0$$

almost surely. Write $P_1 = (p_{11}, \dots, p_{1p})$, it is equivalent to showing

$$\max_{i=1, \dots, p} \left| \frac{p_{1i}^\top \tilde{\Phi}_2 p_{1i} - p_{1i}^\top \Phi p_{1i}}{p_{1i}^\top \Phi p_{1i}} \right| \rightarrow 0$$

almost surely.

We can write

$$\Delta X_\ell = \left(\int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt \right)^{1/2} \Phi^{1/2} Z_\ell, \quad (\text{S1})$$

where the Z_ℓ 's are independent of each other and each Z_ℓ is a p dimensional vector with independent standard normal entries. Then we can decompose

$$\frac{p_{1i}^\top \tilde{\Phi}_2 p_{1i} - p_{1i}^\top \Phi p_{1i}}{p_{1i}^\top \Phi p_{1i}} = I_{i1} + I_{i2} \quad (i = 1, \dots, p),$$

where

$$I_{i1} = \frac{p_{1i}^T \tilde{\Phi}_2 p_{1i} - n_2^{-1} \sum_{\ell \in I_2} (p_{1i}^T \Phi^{1/2} Z_\ell)^2}{p_{1i}^T \Phi p_{1i}}, \quad I_{i2} = \frac{n_2^{-1} \sum_{\ell \in I_2} (p_{1i}^T \Phi^{1/2} Z_\ell)^2 - p_{1i}^T \Phi p_{1i}}{p_{1i}^T \Phi p_{1i}}.$$

Since we can write

$$n_2^{-1} \sum_{\ell \in I_2} (p_{1i}^T \Phi^{1/2} Z_\ell)^2 = p_{1i}^T (n_2^{-1} \sum_{\ell \in I_2} \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2}) p_{1i} \quad (i = 1, \dots, p),$$

and $n_2^{-1} \sum_{\ell \in I_2} \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2}$ is a proper sample covariance matrix for estimating Φ , Lemma 1 of Lam (2016a) implies almost surely,

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$$\max_{i=1, \dots, p} |I_{i2}| \rightarrow 0.$$

Hence it remains to show that $\max_{i=1, \dots, p} |I_{i1}| \rightarrow 0$ almost surely. To this end, consider

$$\begin{aligned} \max_{i=1, \dots, p} |I_{i1}| &= \max_{i=1, \dots, p} \left| \frac{n_2^{-1} \sum_{\ell \in I_2} \frac{(p_{1i}^T \Phi^{1/2} Z_\ell)^2}{Z_\ell^T \Phi Z_\ell / p} - p_{1i}^T (n_2^{-1} \sum_{\ell \in I_2} \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2}) p_{1i}}{p_{1i}^T \Phi p_{1i}} \right| \\ &= \max_{i=1, \dots, p} \left| \frac{n_2^{-1} \sum_{\ell \in I_2} \left(\frac{1}{Z_\ell^T \Phi Z_\ell / p} - 1 \right) p_{1i}^T \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2} p_{1i}}{p_{1i}^T \Phi p_{1i}} \right| \\ &\leq \max_{\ell \in I_2} \left| \frac{1}{Z_\ell^T \Phi Z_\ell / p} - 1 \right| (1 + \max_{i=1, \dots, p} |I_{i2}|) \rightarrow 0 \end{aligned}$$

almost surely, if we can show further that $\max_{\ell \in I_2} |(Z_\ell^T \Phi Z_\ell / p)^{-1} - 1| \rightarrow 0$ almost surely.

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Using Lemma 2.7 of Bai & Silverstein (1998),

$$E\{Z_\ell^T \Phi Z_\ell - \text{tr}(\Phi)\}^6 \leq K_6 \{E^3 |z_{\ell,1}|^4 \text{tr}^3(\Phi^2) + E |z_{\ell,1}|^{12} \text{tr}(\Phi^6)\},$$

where K_6 is a constant independent of ℓ , n and p . This implies that, since $\text{tr}(\Phi) = p$,

$$\begin{aligned} E \left(\max_{\ell \in I_2} \left| \frac{Z_\ell^T \Phi Z_\ell}{p} - 1 \right|^6 \right) &\leq n_2 K_6 \left\{ E^3 |z_{\ell,1}|^4 \frac{\text{tr}^3(\Phi^2)}{p^6} + E |z_{\ell,1}|^{12} \frac{\text{tr}(\Phi^6)}{p^6} \right\} \\ &= O(n_2 p^{-3}). \end{aligned}$$

The rate in the last line comes from Assumption 2 that $\Theta_t \Theta_t^T = \gamma_t^2 \Phi$ has all its eigenvalues uniformly bounded away from 0 and infinity, so that $\text{tr}^3(\Phi^2) = O(p^3)$ and $\text{tr}(\Phi^6) = O(p)$, and the fact that the higher order moments of the $z_{\ell,1}$'s are all finite since they are all normally distributed.

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Finally, since $n_2 = O(n^{1/2})$ and p has the same order as n , we have $O(n_2 p^{-3}) = O(n^{-5/2})$. Since $\sum_{n \geq 1} n^{-5/2} < \infty$, through the Borel-Cantelli lemma, we have proved that $\max_{\ell \in I_2} |Z_\ell^T \Phi Z_\ell / p - 1| \rightarrow 0$ almost surely, meaning that $\max_{\ell \in I_2} |(Z_\ell^T \Phi Z_\ell / p)^{-1} - 1| \rightarrow 0$ almost surely as well. This completes the proof of the lemma. $\square$

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Proof of Theorem 1. Dropping the superscript (j) , by Lemma 1, $\hat{\Phi} = P_1 \text{diag}(P_1^T \tilde{\Phi}_2 P_1) P_1^T$ defined in (5) is almost surely positive definite since all its eigenvalues are almost surely $p_{1i}^T \Phi p_{1i}$ ($i = 1, \dots, p$) as $n, p \rightarrow \infty$ such that $p/n \rightarrow c > 0$, and Assumption 2 ensures that these values are uniformly bounded away from 0 and infinity. Hence, the proof of the theorem completes if we can show that $\text{tr}(\Sigma_p^{\text{RCV}})/p > 0$ uniformly almost surely.

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To this end, consider

$$\begin{aligned}
\left| \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} - \int_0^1 \gamma_t^2 dt \right| &= \left| \frac{\sum_{\ell=1}^n \Delta X_\ell^T \Delta X_\ell}{p} - \int_0^1 \gamma_t^2 dt \right| \\
&= \left| \frac{\sum_{\ell=1}^n \int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt Z_\ell^T \Phi Z_\ell}{p} - \int_0^1 \gamma_t^2 dt \right| \\
&\leq \max_{\ell=1,\dots,n} \left| \frac{Z_\ell^T \Phi Z_\ell}{p} - 1 \right| \int_0^1 \gamma_t^2 dt.
\end{aligned}$$

From the proof of $\max_{\ell \in I_2} |Z_\ell^T \Phi Z_\ell / p - 1| \rightarrow 0$ almost surely in the last part of the proof of Lemma 1, we can replace n_2 there by n and conclude that

$$\begin{aligned}
E \left(\max_{1 \leq \ell \leq n} \left| \frac{Z_\ell^T \Phi Z_\ell}{p} - 1 \right|^6 \right) &\leq n K_6 \left\{ E^3 |z_{\ell,1}|^4 \frac{\text{tr}^3(\Phi^2)}{p^6} + E |z_{\ell,1}|^{12} \frac{\text{tr}(\Phi^6)}{p^6} \right\} \\
&= O(np^{-3}) = O(n^{-2}).
\end{aligned}$$

Since $\sum_{n \geq 1} n^{-2} < \infty$, we can conclude that $\max_{\ell=1,\dots,n} |Z_\ell^T \Phi Z_\ell / p - 1| \rightarrow 0$ almost surely. This shows that $\text{tr}(\Sigma_p^{\text{RCV}})/p \rightarrow \int_0^1 \gamma_t^2 dt$ almost surely, which is uniformly larger than 0 by Assumption 1. This completes the proof of the theorem. $\square$

To prove Theorem 2, we first present the following lemma and its proof. We drop the superscript (j) in the definition of $P_1^{(j)}, \tilde{\Phi}_i^{(j)} (i = 1, 2; j = 1, \dots, M)$ defined in (5) through the lemma, which is true for each $j = 1, \dots, M$.

LEMMA 2. *Let all the assumptions in Lemma 1 hold, together with Assumption 4 and 5. Denote by $v_1^{(1)} \geq \dots \geq v_p^{(1)}$ the eigenvalues of $\tilde{\Phi}_1$ with corresponding eigenvectors $p_{11}, \dots, p_{1p}$, and $v_1 \geq \dots \geq v_p$ the eigenvalues of $\tilde{\Phi}$ defined in (4) with corresponding eigenvectors $p_1, \dots, p_p$. Then there exist positive functions $\delta_1(\cdot) = \delta(\cdot)$ and distribution functions $F_1 = F$ such that almost surely,*

$$\begin{aligned}
p^{-1} \sum_{j=1}^p 1_{\{x \geq v_j^{(1)}\}} &\rightarrow F_1(x), \quad p^{-1} \sum_{j=1}^p 1_{\{x \geq v_j\}} \rightarrow F(x), \\
p^{-1} \sum_{j=1}^p p_{1i}^T \Phi p_{1i} 1_{\{x \geq v_j^{(1)}\}} &\rightarrow \int_{-\infty}^x \delta_1(\lambda) dF_1(\lambda), \quad p^{-1} \sum_{j=1}^p p_i^T \Phi p_i 1_{\{x \geq v_j\}} \rightarrow \int_{-\infty}^x \delta(\lambda) dF(\lambda).
\end{aligned}$$

We do not write down the explicit form of the functions $\delta_1(\cdot)$ and $\delta(\cdot)$ since they are not important for the proof of any subsequent theorems. Interested readers are referred to equation (2.7) and (2.9) of Lam (2016a).

Proof of Lemma 2. Write $\Delta X_\ell = (\int_{\tau_{n,\ell-1}}^{\tau_{n,\ell}} \gamma_t^2 dt)^{1/2} \Phi^{1/2} Z_\ell$ as in (S1). Define

$$\tilde{\Phi}_{\text{sam}} = n^{-1} \sum_{\ell=1}^n \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2}, \quad \tilde{\Phi}_{i,\text{sam}} = n_i^{-1} \sum_{\ell \in I_i} \Phi^{1/2} Z_\ell Z_\ell^T \Phi^{1/2} \quad (i = 1, 2),$$

which are all proper sample covariance matrices. Let $v_{1,\text{sam}}^{(i)} \geq \dots \geq v_{p,\text{sam}}^{(i)}$ be the eigenvalues of $\tilde{\Phi}_{i,\text{sam}}$ with corresponding eigenvectors $p_{i1,\text{sam}}, \dots, p_{ip,\text{sam}}$. Also, let $v_{1,\text{sam}} \geq \dots \geq v_{p,\text{sam}}$ be the eigenvalues of $\tilde{\Phi}_{\text{sam}}$ with corresponding eigenvectors $p_{1,\text{sam}}, \dots, p_{p,\text{sam}}$. Suppose we are able to

show the following. Almost surely,

$$\begin{aligned}
p^{-1}\text{tr}\{(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}\} - p^{-1}\text{tr}\{(\tilde{\Phi}_1 - zI_p)^{-1}\} &\rightarrow 0 \quad (z \in \mathbb{C}^+), \\
p^{-1}\text{tr}\{(\check{\Phi}_{\text{sam}} - zI_p)^{-1}\} - p^{-1}\text{tr}\{(\check{\Phi} - zI_p)^{-1}\} &\rightarrow 0 \quad (z \in \mathbb{C}^+), \\
p^{-1}\text{tr}\{(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}\Phi\} - p^{-1}\text{tr}\{(\tilde{\Phi}_1 - zI_p)^{-1}\Phi\} &\rightarrow 0 \quad (z \in \mathbb{C}^+), \\
p^{-1}\text{tr}\{(\check{\Phi}_{\text{sam}} - zI_p)^{-1}\Phi\} - p^{-1}\text{tr}\{(\check{\Phi} - zI_p)^{-1}\Phi\} &\rightarrow 0 \quad (z \in \mathbb{C}^+).
\end{aligned} \tag{S2}$$

The left hand side of the above can in fact be written as differences of Stieltjes transforms of certain nondecreasing functions. The differences in their inverse Stieltjes transforms must then converge to 0 almost surely as well, i.e., at the point of continuity x of these nondecreasing functions, almost surely,

$$\begin{aligned}
p^{-1} \sum_{j=1}^p 1_{\{x \geq v_{j,\text{sam}}^{(1)}\}} - p^{-1} \sum_{j=1}^p 1_{\{x \geq v_j^{(1)}\}} &\rightarrow 0, \\
p^{-1} \sum_{j=1}^p 1_{\{x \geq v_{j,\text{sam}}\}} - p^{-1} \sum_{j=1}^p 1_{\{x \geq v_j\}} &\rightarrow 0, \\
p^{-1} \sum_{j=1}^p p_{1j,\text{sam}}^T \Phi p_{1j,\text{sam}} 1_{\{x \geq v_{j,\text{sam}}^{(1)}\}} - p^{-1} \sum_{j=1}^p p_{1j}^T \Phi p_{1j} 1_{\{x \geq v_j^{(1)}\}} &\rightarrow 0, \\
p^{-1} \sum_{j=1}^p p_{j,\text{sam}}^T \Phi p_{j,\text{sam}} 1_{\{x \geq v_{j,\text{sam}}\}} - p^{-1} \sum_{j=1}^p p_j^T \Phi p_j 1_{\{x \geq v_j\}} &\rightarrow 0.
\end{aligned} \tag{S3}$$

Interested readers are referred to Lam (2016a) or Ledoit & P  ch   (2011) for the definitions of Stieltjes transform and its inversion. Theorem 4 of Ledoit & P  ch   (2011) indicates that for all $x \in \mathbb{R}$, there exist functions $\delta_1(\cdot)$ and $\delta(\cdot)$ with corresponding distribution functions F_1 and F such that almost surely,

$$\begin{aligned}
p^{-1} \sum_{j=1}^p 1_{\{\lambda \geq v_{j,\text{sam}}^{(1)}\}} &\rightarrow F_1(\lambda), \\
p^{-1} \sum_{j=1}^p 1_{\{\lambda \geq v_{j,\text{sam}}\}} &\rightarrow F(\lambda), \\
p^{-1} \sum_{j=1}^p p_{1j,\text{sam}}^T \Phi p_{1j,\text{sam}} 1_{\{x \geq v_{j,\text{sam}}^{(1)}\}} &\rightarrow \int_{-\infty}^x \delta_1(\lambda) dF(\lambda), \\
p^{-1} \sum_{j=1}^p p_{j,\text{sam}}^T \Phi p_{j,\text{sam}} 1_{\{x \geq v_{j,\text{sam}}\}} &\rightarrow \int_{-\infty}^x \delta(\lambda) dF(\lambda).
\end{aligned}$$

At the same time, since $p/n_1, p/n \rightarrow c > 0$, Theorem 4.1 of Bai & Silverstein (2010) says that the two limits F_1 and F are equal, and since $\delta_1(\cdot)$ and $\delta(\cdot)$ depend on F_1 and F and on the same $c > 0$, we must have $\delta_1(\cdot) = \delta(\cdot)$ also.

With the above, (S3) immediately implies the results we need. Hence it remains to show (S2).

To prove the first and third results of (S2), consider

$$\begin{aligned}
& p^{-1} \text{tr}\{(\tilde{\Phi}_1 - zI_p)^{-1} \Phi^k\} \\
&= p^{-1} \text{tr}\{[I_p - (\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1)]^{-1}(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1} \Phi^k\} \\
&= p^{-1} \text{tr}\{(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1} \Phi^k\} + R \quad (k = 0, 1),
\end{aligned}$$

where

$$R = \sum_{j \geq 1} p^{-1} \text{tr}\{[(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1)]^j (\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1} \Phi^k\} \quad (k = 0, 1).$$

The term R comes from a Neumann series expansion, which is valid since almost surely

$$\begin{aligned}
r &= \|(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}(\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1)\| \leq \|(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}\| \|\tilde{\Phi}_{1,\text{sam}} - \tilde{\Phi}_1\| \\
&\leq \frac{1}{|\text{Im}(z)|} \max_{\ell \in I_1} \left| \frac{1}{Z_\ell^T \Phi Z_\ell / p} - 1 \right| \|\tilde{\Phi}_{1,\text{sam}}\| \\
&\leq \frac{\|\Phi^{1/2}\|^2}{|\text{Im}(z)|} \max_{\ell \in I_1} \left| \frac{1}{Z_\ell^T \Phi Z_\ell / p} - 1 \right| \left\| n_1^{-1} \sum_{\ell \in I_1} Z_\ell Z_\ell^T \right\| \\
&\leq \frac{\|\Phi^{1/2}\|^2 (1 + \sqrt{c})^2}{|\text{Im}(z)|} \max_{\ell \in I_1} \left| \frac{1}{Z_\ell^T \Phi Z_\ell / p} - 1 \right| \rightarrow 0,
\end{aligned}$$

where we used Lemma S.2 of Lam (2016b) to conclude $\|(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}\| \leq 1/|\text{Im}(z)|$. We used Theorem 5.11 of Bai & Silverstein (2010) for $\|n_1^{-1} \sum_{\ell \in I_1} Z_\ell Z_\ell^T\| \leq (1 + \sqrt{c})^2$ almost surely, since $p/n_1 \rightarrow c > 0$. Finally, the term $\max_{\ell \in I_1} |(Z_\ell^T \Phi Z_\ell / p)^{-1} - 1| \rightarrow 0$ almost surely by the last part of the proof of Lemma 1. Hence almost surely,

$$|R| \leq \sum_{j \geq 1} r^j \|\Phi^k\| \|(\tilde{\Phi}_{1,\text{sam}} - zI_p)^{-1}\| \leq \frac{r \|\Phi^k\|}{(1 - r)|\text{Im}(z)|} \rightarrow 0,$$

so that we have proved the first and third results in (S2). For the other two results, the proof follows exactly the same lines as before after replacing $\tilde{\Phi}_{1,\text{sam}}$ by $\check{\Phi}_{\text{sam}}$ and $\tilde{\Phi}_1$ by $\check{\Phi}$. This completes the proof of the lemma. $\square$

Proof of Theorem 2. Observe that

$$\hat{\Sigma}_{m,M} = \frac{1}{M} \sum_{j=1}^M \hat{\Sigma}_m^{(j)}, \quad \hat{\Sigma}_m^{(j)} = \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} \hat{\Phi}^{(j)}.$$

If we can prove that $\text{EffLoss}(\Sigma_p, \hat{\Sigma}_m^{(j)}) \leq 0$ almost surely, then we can follow exactly the same lines in the proof of Theorem 6 in Lam (2016a) to complete the proof Theorem 2.

Since M is finite, we drop the superscript (j) in $\hat{\Sigma}_m^{(j)}$, and define

$$\hat{\Sigma}_m = \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} \hat{\Phi} = \frac{\text{tr}(\Sigma_p^{\text{RCV}})}{p} P_1 \text{diag}(P_1^T \tilde{\Phi}_2 P_1) P_1^T.$$

It remains to show that $\text{EffLoss}(\Sigma_p, \hat{\Sigma}_m) \leq 0$. Consider the Frobenius loss first. Define $\theta = \int_0^1 \gamma_t^2 dt$, $\hat{\theta} = \text{tr}(\Sigma_p^{\text{RCV}})/p$ and $\Phi_{\text{Ideal}} = P \text{diag}(P^T \Phi P) P^T$. Then

$$\text{EffLoss}(\Sigma_p, \hat{\Sigma}_m) \leq 1 - \left\{ \frac{p^{-1/2} \|(\hat{\theta} - \theta) \hat{\Phi}\|_F}{p^{-1/2} \theta \|\Phi_{\text{Ideal}} - \Phi\|_F} + \frac{p^{-1/2} \|\hat{\Phi} - \Phi\|_F}{p^{-1/2} \|\Phi_{\text{Ideal}} - \Phi\|_F} \right\}^{-2}, \quad (\text{S4})$$

where $p^{-1/2} \|(\hat{\theta} - \theta) \hat{\Phi}\|_F \leq |\hat{\theta} - \theta| \max_{i=1, \dots, p} p_{1i}^T \tilde{\Phi}_2 p_{1i} \rightarrow 0$ almost surely by Lemma 1 and the proof of Theorem 1. Since we are equivalently assuming that $p^{-1/2} \|\Phi_{\text{Ideal}} - \Phi\|_F \not\rightarrow 0$, it remains to show that $p^{-1/2} \|\hat{\Phi} - \Phi\|_F / (p^{-1/2} \|\Phi_{\text{Ideal}} - \Phi\|_F) \rightarrow 1$ almost surely. Observe that

$$\frac{p^{-1} \|\hat{\Phi} - \Phi\|_F^2}{p^{-1} \|\Phi_{\text{Ideal}} - \Phi\|_F^2} = \frac{p^{-1} \sum_{i=1}^p (p_{1i}^T \tilde{\Phi}_2 p_{1i} - p_{1i}^T \Phi p_{1i})^2}{p^{-1} \|P \text{diag}(P^T \Phi P) P^T - \Phi\|_F^2} + \frac{p^{-1} \|P_1 \text{diag}(P_1^T \Phi P_1) P_1^T - \Phi\|_F^2}{p^{-1} \|P \text{diag}(P^T \Phi P) P^T - \Phi\|_F^2}.$$

By Assumptions 4 and 5, and the results of Lemma 2, almost surely,

$$\begin{aligned} p^{-1} \|P_1 \text{diag}(P_1^T \Phi P_1) P_1^T - \Phi\|_F^2 &= p^{-1} \text{tr}(\Phi^2) - p^{-1} \sum_{i=1}^p (p_{1i}^T \Phi p_{1i})^2 \\ &\rightarrow \int \tau^2 dH(\tau) - \int \delta_1^2(\lambda) dF_1(\lambda) \\ &= \int \tau^2 dH(\tau) - \int \delta^2(\lambda) dF_s(\lambda), \end{aligned}$$

which is non-zero if $\Phi \neq I_p$. This is also the almost sure limit of $p^{-1} \|\Phi_{\text{Ideal}} - \Phi\|_F^2$, and hence $p^{-1/2} \|\hat{\Phi} - \Phi\|_F / (p^{-1/2} \|\Phi_{\text{Ideal}} - \Phi\|_F) \rightarrow 1$ almost surely if we can also show that $p^{-1} \sum_{i=1}^p (p_{1i}^T \tilde{\Phi}_2 p_{1i} - p_{1i}^T \Phi p_{1i})^2 \rightarrow 0$ almost surely. By Lemma 1, almost surely,

$$p^{-1} \sum_{i=1}^p (p_{1i}^T \tilde{\Phi}_2 p_{1i} - p_{1i}^T \Phi p_{1i})^2 \leq \max_{i=1, \dots, p} \left| \frac{p_{1i}^T \tilde{\Phi}_2 p_{1i} - p_{1i}^T \Phi p_{1i}}{p_{1i}^T \Phi p_{1i}} \right| \max_{i=1, \dots, p} p_{1i}^T \Phi p_{1i} \rightarrow 0.$$

This completes the proof for the Frobenius loss.

For the inverse Stein loss, by Lemma 2, almost surely,

$$\begin{aligned} p^{-1} L(\Sigma_p, \Sigma_{\text{Ideal}}) &= p^{-1} \sum_{i=1}^p \log(p_i^T \Phi p_i) - p^{-1} \sum_{i=1}^p \log(v_{n,i}) \\ &\rightarrow \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau), \end{aligned}$$

160 where $v_{n,i}$ is the i th largest eigenvalue of Φ . Now consider the decomposition

$$\begin{aligned}
 p^{-1}L(\Sigma_p, \widehat{\Sigma}_p) &= I_1 + I_2 + I_3 + I_4 + I_5, \\
 I_1 &= \log(\widehat{\theta}/\theta), \\
 I_2 &= \left(\frac{\theta}{\widehat{\theta}} - 1\right)p^{-1} \sum_{i=1}^p \left(\frac{p_{1i}^T \Phi p_{1i}}{p_{1i}^T \widetilde{\Phi} p_{1i}} \right), \\
 I_3 &= p^{-1} \sum_{i=1}^p \left(\frac{p_{1i}^T \Phi p_{1i}}{p_{1i}^T \widetilde{\Phi} p_{1i}} \right) - 1, \\
 I_4 &= p^{-1} \sum_{i=1}^p \log \left(\frac{p_{1i}^T \Phi p_{1i}}{p_{1i}^T \widetilde{\Phi} p_{1i}} \right), \\
 I_5 &= p^{-1} \sum_{i=1}^p \log(p_{1i}^T \Phi p_{1i}) - p^{-1} \sum_{i=1}^p \log(v_{n,i}).
 \end{aligned}$$

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We can prove that $I_1, I_2, I_3, I_4 \rightarrow 0$ almost surely by the proof of Theorem 1, and the result of Lemma 1. By Lemma 2, we can show that almost surely,

$$I_5 \rightarrow \int \log(\delta(\lambda))dF(\lambda) - \int \log(\tau)dH(\tau),$$

170 so that $p^{-1}L(\Sigma_p, \Sigma_{\text{Ideal}})/\{p^{-1}L(\Sigma_p, \widehat{\Sigma}_p)\} \rightarrow 1$ almost surely, showing $\text{EffLoss}(\Sigma_p, \widehat{\Sigma}_m) \rightarrow 0$ almost surely. This completes the proof of the theorem. $\square$

Proof of Theorem 3. For a matrix $A = (a_{ij})$, define $\|A\|_1 = \max_j \sum_{i \geq 1} |a_{ij}|$ and $\|A\|_\infty = \max_i \sum_{j \geq 1} |a_{ij}|$. Then almost surely,

$$\begin{aligned}
 p^{1/2} \|\widehat{w}_{\text{opt}}\|_\infty &\leq \frac{p^{1/2} \|\widehat{\Sigma}_p^{-1}\|_1}{1_p^T \widehat{\Sigma}_p^{-1} 1_p} = \frac{p^{1/2} \|\widehat{\Phi}^{-1}\|_\infty}{1_p^T \widehat{\Phi}^{-1} 1_p} \leq \frac{p^{1/2} \{p^{1/2} \lambda_{\min}^{-1}(\widehat{\Phi})\}}{p \lambda_{\min}(\widehat{\Phi}^{-1})} \\
 &= \frac{\lambda_{\max}(\widehat{\Phi})}{\lambda_{\min}(\widehat{\Phi})} \leq \frac{\max_{i=1, \dots, p} p_{1i}^T \widetilde{\Phi} p_{1i}}{\min_{i=1, \dots, p} p_{1i}^T \widetilde{\Phi} p_{1i}} \rightarrow \frac{\max_{i=1, \dots, p} p_{1i}^T \Phi p_{1i}}{\min_{i=1, \dots, p} p_{1i}^T \Phi p_{1i}} \\
 &\leq \frac{\lambda_{\max}(\Phi)}{\lambda_{\min}(\Phi)} = \text{Cond}(\Phi),
 \end{aligned}$$

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where we used the results of Lemma 1 for the almost sure convergence. For the theoretical minimum variance portfolio w_{opt} ,

$$\begin{aligned}
 p^{1/2} \|w_{\text{opt}}\|_\infty &\leq \frac{p^{1/2} \|\Sigma_p^{-1}\|_1}{1_p^T \Sigma_p^{-1} 1_p} = \frac{p^{1/2} \|\Phi^{-1}\|_\infty}{1_p^T \Phi^{-1} 1_p} \leq \frac{p^{1/2} \{p^{1/2} \lambda_{\min}^{-1}(\Phi)\}}{p \lambda_{\min}(\Phi^{-1})} \\
 &= \frac{\lambda_{\max}(\Phi)}{\lambda_{\min}(\Phi)} = \text{Cond}(\Phi).
 \end{aligned}$$

For the actual risk bound, almost surely,

$$\begin{aligned}
 pR^2(\hat{w}_{\text{opt}}) &= \frac{p \mathbf{1}_p^T \hat{\Sigma}_p^{-1} \Sigma_p \hat{\Sigma}_p^{-1} \mathbf{1}_p}{(\mathbf{1}_p^T \hat{\Sigma}_p^{-1} \mathbf{1}_p)^2} \leq \frac{p \{p \lambda_{\max}^2(\hat{\Sigma}_p^{-1}) \lambda_{\max}(\Sigma_p)\}}{p^2 \lambda_{\min}^2(\hat{\Sigma}_p^{-1})} = \left\{ \frac{\lambda_{\max}(\hat{\Phi})}{\lambda_{\min}(\hat{\Phi})} \right\}^2 \lambda_{\max}(\Sigma_p) \\
 &= \left(\frac{\max_{i=1, \dots, p} p \mathbf{1}_i^T \tilde{\Phi} \mathbf{1}_i}{\min_{i=1, \dots, p} p \mathbf{1}_i^T \tilde{\Phi} \mathbf{1}_i} \right)^2 \lambda_{\max}(\Sigma_p) \rightarrow \left(\frac{\max_{i=1, \dots, p} p \mathbf{1}_i^T \Phi \mathbf{1}_i}{\min_{i=1, \dots, p} p \mathbf{1}_i^T \Phi \mathbf{1}_i} \right)^2 \lambda_{\max}(\Sigma_p) \\
 &\leq \left\{ \frac{\lambda_{\max}(\Phi)}{\lambda_{\min}(\Phi)} \right\}^2 \lambda_{\max}(\Sigma_p) = \text{Cond}^2(\Phi) \lambda_{\max}(\Sigma_p).
 \end{aligned}$$

Finally, for the theoretical minimum variance portfolio w_{opt} ,

$$pR^2(w_{\text{opt}}) = \frac{p \mathbf{1}_p^T \Sigma_p^{-1} \Sigma_p \Sigma_p^{-1} \mathbf{1}_p}{(\mathbf{1}_p^T \Sigma_p^{-1} \mathbf{1}_p)^2} = \frac{p}{\mathbf{1}_p^T \Sigma_p^{-1} \mathbf{1}_p} \leq \lambda_{\max}(\Sigma_p).$$

This completes the proof of the theorem. $\square$

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